Time: 90 Minutes
Maximum Marks: 100

General Instructions:

1. This paper is divided into four sections: A, B, C, and D.
2. **Section A** consists of 25 Multiple Choice Questions (MCQs). Each question carries 2 marks. (25 × 2 = 50 marks)
3. **Section B** consists of 10 True/False and Fill in the Blanks questions. Each question carries 1 mark. (10 × 1 = 10 marks)
4. **Section C** consists of 5 Matching Type questions. Each question carries 2 marks. (5 × 2 = 10 marks)
5. **Section D** consists of 10 Short Answer and Case-Based questions. Each question carries 3 marks. (10 × 3 = 30 marks)
6. All questions are compulsory. Write neatly and number your answers correctly.

Section A: Multiple Choice Questions (25 × 2 = 50 Marks)

1. The difference between the greatest 5-digit number and the smallest 4-digit number is:
a) 98999
b) 98998
c) 99999
d) 89999

2. How many numbers between 100 and 200 are divisible by both 3 and 5?
a) 5
b) 6
c) 7
d) 8

3. The sum of all factors of 28 is:
a) 28
b) 42
c) 56
d) 64

4. The HCF of two numbers is 12 and their LCM is 144. If one number is 36, the other number is:
a) 24
b) 36
c) 48
d) 60

5. What is the remainder when 2ˆ100 is divided by 5?
a) 0
b) 1
c) 2
d) 4

6. The product of the first 5 prime numbers is:
a) 2310
b) 2100
c) 2312
d) 2200

7. Which of the following fractions is the smallest?
a) 3/8
b) 5/12
c) 7/16
d) 9/20

8. A rope of length 2/3 m is cut into pieces of length 1/6 m each. How many pieces are obtained?
a) 3
b) 4
c) 5
d) 6

9. The value of 3.125 × 8.4 is:
a) 26.25
b) 26.250
c) 26.5
d) 26.75

10. If 2/3 of a number is 48, what is 5/4 of the same number?
a) 60
b) 75
c) 80
d) 90

11. A square and a rectangle have the same perimeter. If the side of the square is 12 cm and the length of the rectangle is 16 cm, the breadth of the rectangle is:
a) 6 cm
b) 8 cm
c) 10 cm
d) 12 cm

12. The area of a triangle with base 18 cm and height 12 cm is:
a) 108 cm^2
b) 216 cm^2
c) 54 cm^2
d) 120 cm^2

13. In a triangle ABC, $\angle A = 65°$ and $\angle B = 55°$. The measure of $\angle C$ is:
a) 50°
b) 55°
c) 60°
d) 65°

14. How many diagonals does a regular heptagon have?
a) 7
b) 10
c) 14
d) 21

15. If the ratio of boys to girls in a class is 5:7 and there are 36 students in total, the number of boys is:
a) 15
b) 20
c) 25
d) 30

16. If x:y = 2:3 and y:z = 4:5, then x:z is:
a) 8:15
b) 2:5
c) 4:15
d) 6:15

17. A train travels at a speed of 75 km/h. How far will it travel in 2 hours 40 minutes?
a) 180 km
b) 190 km
c) 200 km
d) 210 km

18. The average of 8 numbers is 45. If one number is removed, the average becomes 43. The removed number is:
a) 53
b) 56
c) 59
d) 61

19. Solve for x: $3x - 7 = 2x + 13$
a) $x = 6$
b) $x = 10$
c) $x = 20$
d) $x = 26$

20. The next term in the sequence 3, 7, 13, 21, 31, __ is:
a) 41
b) 43
c) 45
d) 47

21. The sum of the first 20 even natural numbers is:
a) 400
b) 410
c) 420
d) 440

22. A number when divided by 13 leaves a remainder of 7. If twice the same number is divided by 13, the remainder is:
a) 1
b) 7
c) 14
d) 0

23. What is the digit in the units place of 7ˆ53?
a) 1
b) 3
c) 7
d) 9

24. The circumference of a circle with radius 14 cm is: (Take π = 22/7)
a) 44 cm
b) 66 cm
c) 88 cm
d) 96 cm

25. If the area of a circle is 154 cm^2, its radius is: (Take π = 22/7)
a) 7 cm
b) 14 cm
c) 21 cm
d) 49 cm

Section B: True/False and Fill in the Blanks (10 × 1 = 10 Marks)

A. True or False (Write 'True' or 'False')

26. The sum of three consecutive integers is always divisible by 3.

27. The number 1 is a prime number.

28. A circle has infinite lines of symmetry.

29. The product of two negative integers is always positive.

30. The perimeter of a circle is also called its circumference.

B. Fill in the Blanks (Write the complete sentence with the correct word)

31. The value of π (pi) up to two decimal places is __________.

32. The number of faces of a cube is __________.

33. A quadrilateral with all sides equal but no right angles is called a __________.

34. The unit of measurement for angles is __________.

35. The smallest composite number is __________.

Section C: Matching Type Questions (5 × 2 = 10 Marks)

Match Column A with Column B. Write the correct letter next to the number.

Column A	Column B
36. Triangle	a. 360°
37. Quadrilateral	b. 180°
38. Pentagon	c. 720°
39. Hexagon	d. 540°
40. Circle	e. 360° at center

Answers format: 36-b, 37-a, etc.

Section D: Short Answer and Case-Based Questions (10 × 3 = 30 Marks)

41. The Number Theory Challenge (Case-Based)

A student is investigating properties of numbers.

a) Find the HCF and LCM of 48, 72, and 108 using prime factorization. (2 marks)
b) Verify that HCF × LCM ≠ Product of the three numbers. Give a reason. (1 mark)

42. The Fraction and Decimal Problem (Application-Based)

A tank has a capacity of 15.75 litres. It is filled with 8.5 litres of water.

a) How much more water can be added to the tank? (1.5 marks)
b) Express the amount of water already in the tank as a fraction of the total capacity in its simplest form. (1.5 marks)

43. The Geometry Challenge (Experimental Design)

A rectangular garden has a length of 24 m and a breadth of 18 m. A path of uniform width 2 m runs around the outside of the garden.

a) Find the area of the garden. (1 mark)
b) Find the dimensions of the garden including the path. (1 mark)
c) Find the area of the path only. (1 mark)

44. The Ratio and Proportion Problem (Conceptual Understanding)

The ratio of the ages of Aisha and Rahul is 5:7. After 6 years, the ratio of their ages will be 7:9.

a) Let the present ages be 5x and 7x. Form an equation after 6 years. (1.5 marks)
b) Find their present ages. (1.5 marks)

45. The Speed, Distance, Time Problem (Logical Reasoning)

A cyclist travels from Town X to Town Y at a speed of 15 km/h and returns from Town Y to Town X at a speed of 10 km/h. The total time taken for the round trip is 5 hours.

a) Find the distance between Town X and Town Y. (2 marks)
b) What is the average speed of the cyclist for the entire round trip? (1 mark)

46. The Pattern Recognition Question (Application)

Observe the following pattern:
$1^2 = 1$
$2^2 = 1 + 3$
$3^2 = 1 + 3 + 5$
$4^2 = 1 + 3 + 5 + 7$

a) Write the expression for 5^2 using the same pattern. (1 mark)
b) What is the sum of the first 10 odd numbers? (1 mark)
c) What does this pattern demonstrate about square numbers? (1 mark)

47. The Mensuration Problem (Conceptual Understanding)

A wire of length 44 cm is first bent to form a square and then bent to form a circle.

a) Find the area of the square formed. (1.5 marks)
b) Find the area of the circle formed. (1.5 marks)

48. The Divisibility and Remainder Challenge (Application)

a) Find the smallest number which when divided by 12, 16, and 24 leaves a remainder of 7 in each case. (2 marks)
b) What is the least number that must be added to 4321 to make it divisible by 9? (1 mark)

49. The Average Problem (Conceptual Understanding)

The average of 5 numbers is 40. The average of the first 3 numbers is 35, and the average of the last 2 numbers is 45.

a) Find the sum of all 5 numbers. (1 mark)
b) Find the sum of the first 3 numbers. (1 mark)
c) Find the fourth and fifth numbers if the fourth number is 10 more than the fifth number. (1 mark)

50. The Logical Reasoning Challenge (Higher Order Thinking)

In a class of 50 students, 25 students like Mathematics, 30 students like Science, and 10 students like both subjects.

a) How many students like only Mathematics? (1 mark)
b) How many students like only Science? (1 mark)
c) How many students like neither Mathematics nor Science? (1 mark)